

CF-JEPA: Improving Robustness of JEPA World Models via Controllability Factorization

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Abstract—Controlling an agent with vision requires being able to separate useful information from irrelevant background information. JEPA-style latent world models seem like a natural approach for this, as they do not perform pixel-level reconstruction; however, they are still sensitive to these distractor signals and experience latent collapse. In this work, we introduce Controllability Factorized JEPA (CF-JEPA), a JEPA-style world model which splits the latent space into controllable and uncontrollable subspaces. This factorization allows us to capture all the distractor information into the uncontrollable region, while we use the control-relevant latent information for our task. With this, we show comparable performance across 2D and 3D control tasks under nominal conditions and improved performance under distracted conditions, where CF-JEPA is the only model that does not experience latent collapse. We also validate our model under distracted conditions for a simulated robot task, highlighting the practical application of such a scheme.

Index Terms—World models, JEPA, Visual Control

I. INTRODUCTION

Humans are capable of performing tasks using visual control, regardless of what sort of background visual distractor information or noise might be present. Random objects moving into the visual field can be easily ignored and do not interfere with task progress or break our understanding of the world. Robots deployed in diverse environments should likewise accomplish their tasks despite such visual distractors. Thus, a reasonable expectation of a generative world model might be to have a similar invariant nature to these kinds of external visual stimuli.

However, typical world models [1], [2] struggle under these conditions as they rely on pixel-level reconstruction to learn their representations, and reconstructing such random content is difficult and wasteful. There have been attempts to handle this [3], [4], [5], but they often still rely on reconstructing the scene along with additional techniques or rely on assumptions that may not hold in realistic robotic scenarios.

Joint-Embedding Predictive Architectures (JEPA) appear to be a natural way to handle this kind of distraction, as they remove the generative model built on pixel-level reconstruction and predict dynamics in latent space instead. However, we find that such JEPA world models [6] are still very susceptible to task-irrelevant visual distractor signals, and the latent space collapses as a result. Sensorimotor World Model (SMWM) [7] further adopts an inverse dynamics loss to improve robustness, but it still mixes controllable and uncontrollable information in a single latent.

Building on the JEPA framework, we propose CF-JEPA, a world model with additional latent structure for improved robustness under visual distractor signals. We incorporate controllability factorization, where we partition the latent space into a control-relevant, action-conditioned region and

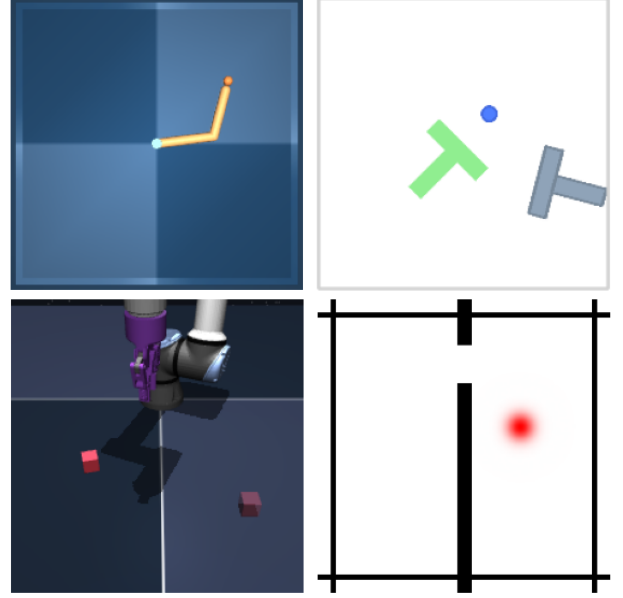


Fig. 1: Primary tasks for evaluation. We test on the DM Control Suite Reacher, PushT, OGBench Cube, and TwoRoom tasks.

a control-irrelevant, action-free region. This explicit partition has a minimal to moderate impact on nominal performance and offers a strong benefit in the presence of visual distractors, where CF-JEPA is the only tested model that does not collapse.

We evaluate CF-JEPA against LeWM and SMWM on four visual control tasks and a simulated robot task. With two visual distractors, LeWM collapses on all tasks, and SMWM collapses on Reacher, where its success rate drops to 0.10 while CF-JEPA retains 0.48. As the number of distractors grows, SMWM degrades rapidly, and its success rate on PushT falls from 0.86 to 0.36 at 16 distractors, whereas CF-JEPA maintains 0.76. We further hypothesize that the benefit of controllability factorization is most evident when the action space requires perceiving the state of the agent, and we confirm this on the simulated robot task. Finally, a diagnostic decoder on Reacher shows that the controllable latent of CF-JEPA preserves the agent, whereas the baseline latents lose the agent.

Our key contributions are the following:

- 1) CF-JEPA, a simple world model architecture that improves robustness to irrelevant visual distractors via controllability factorization, where we separate the irrelevant, uncontrollable environment features from the relevant, controllable features in our latent space.
- 2) We thoroughly compare against benchmarks and show that we match under nominal conditions and outperform under visual perturbations, across a suite of 2D and 3D control tasks.

- 3) We analyze the latent space to examine the effect of the controllability factorization.

II. RELATED WORK

A. World Models

World models [8] are a way of learning the dynamics of an environment via generative models directly from data. Learning the latent representation via reconstruction of the input pixels is a common method, seen in the Dreamer [1], [9], [10] series of models, along with many others including [2], [5], [11], [12], [13]. Other methods for learning models without reconstruction include contrastive learning [3], [14] and prototypical representations [15]. More recently, joint-embedding predictive architectures (JEPA) [16] have gained prominence by constructing the model via predicting feature-space targets rather than pixels. JEPA style models vary widely and include many methods for preventing collapse including using a pretrained encoder as in Dino-WM [17], via a VICReg loss [18] as in PLDM [19], and via a SIGReg [20] loss as in LeWM [6]. We build off LeWM for our work, where we add an inverse dynamics loss and augment the latent space with additional structure in order to improve robustness.

B. Inverse Dynamics for World Models

Inverse dynamics is the process of computing the required forces or torques to produce a given reference trajectory. It is popularly used in robotics for control of manipulator arms [21], [22] and walking robots [23], [24], [25] as well as used in guidance application [26]. In machine learning, it is leveraged for computing actions given a reference video [27], [28] or motion capture sequence [29] and for representation learning [30], [31], [32]. For world modeling specifically, learning an inverse dynamics model is commonly used for acting in imagination, as in [1], [33], as well as representation learning, as in [7], [34], [35]. The most relevant case for our work is the concurrent work Sensorimotor World Models (SMWM) [7], which builds off LeWM, removing the SIGReg loss and adding an inverse model as an anti-collapse mechanism. We also leverage an inverse dynamics loss, but we use it to structure the latent space into controllable and uncontrollable subspaces, giving improved results in terms of robustness over the standard inverse dynamics loss.

C. Distractor Robustness

Rejecting irrelevant distractor signals and focusing on only task relevant information is key for effective visual control. The reconstruction free world model methods above tackle this by not spending capacity modeling every pixel but capturing the underlying state [3], [15]. In model-free RL, bisimulation [4], [36] is one method for generating useful, robust representations from pixel based environments without reconstruction. Task Informed Abstractions [5] and Denoised MDPs [37] partition the latent space similarly to ours, but still require reconstruction and thus still expend some model capacity for representing the pixel level view of the environment, which can be especially difficult if there are

distractor signals. We avoid pixel level reconstruction and gain robustness via separating the controllable latent subspace from the uncontrollable portion, and using only the controllable section for control.

III. METHOD

A. Problem Definition

We study control from pixels without reward data. For each task, we assume access to an offline dataset $\mathcal{D} = \{(x_t, a_t, x_{t+1})\}$ of transitions, where x_t is an RGB image and a_t is the continuous action that maps from x_t to x_{t+1} . The trajectories carry no reward information or task label, and we do not require an optimal policy for generation. Sufficient coverage for accurate representation of the environment dynamics is the primary requirement. From this data, we learn an encoder f that maps the input image x_t to a latent vector z_t as well as a dynamics model that predicts the next latent z_{t+1} using the current latent z_t and action a_t . At inference, the agent receives a goal image x_{goal} , plans in the latent space using model-predictive control via our world model, and we register the success rate as the percentage of trials that successfully get within a threshold of the goal.

We prescribe a factorized structure for the environment MDP dynamics. We split the underlying state s into a controllable part s^c and an uncontrollable part s^u . The action input influences only the controllable section, with the uncontrollable section dynamics progressing autonomously, entirely separate from the controllable dynamics. The agent has no direct access to either state, and only sees the image rendered from them:

$$\begin{aligned} T(s' | s, a) &= T_c(s'^c | s^c, a) T_u(s'^u | s^u), \\ x_t &= q(s_t^c, s_t^u). \end{aligned} \quad (1)$$

Here q is the emission function that renders a state s into an image x . This structure of a factorized state paired with a joint emission is called an exogenous block MDP [38]. Note, this is a strict generalization of the standard MDP format and, if $T_u = I$, $s^u = \emptyset$, and $s^c = s$, would reduce to a standard MDP.

We further assume that visual distractors are captured in the uncontrollable s^u . A distractor signal enters the image through q , it follows its own dynamics T_u , and no action of the agent can change it. Movement in the image background is a distractor, while the arm controlled by the agent is not.

Our goal is to learn a world model that is robust against variation in s^u , such that we maintain high quality control performance even when the uncontrollable state changes. Under ideal circumstances, the factorization perfectly captures the control relevant information and rejects all other information. In reality, it works well for things like extraneous background pixels or noise, but would be incapable of capturing things that directly interfere with the task, like occlusion of the task itself.

B. Background

LeWM [6] learns a latent world model end to end from pixels. It has two parts, an encoder and a predictor,

$$z_t = \text{enc}_\theta(x_t), \quad \hat{z}_{t+1} = \text{pred}_\varphi(z_t, a_t), \quad (2)$$

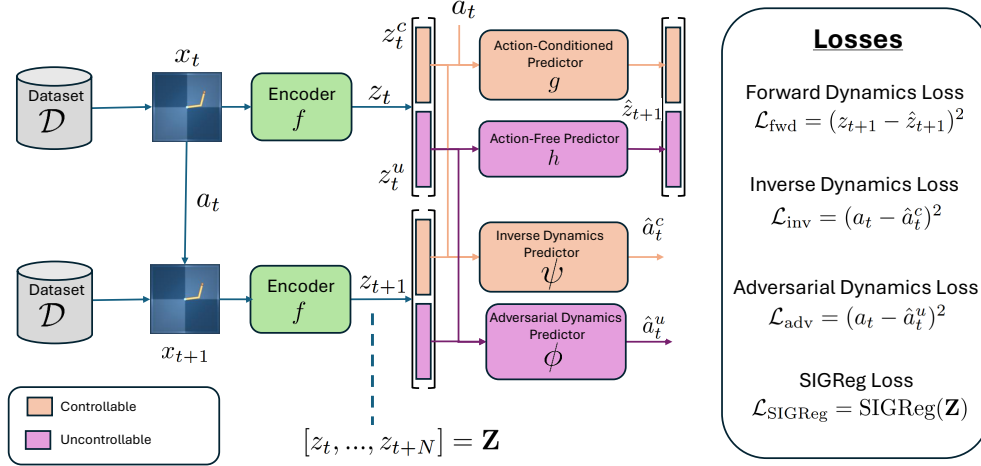


Fig. 2: Overview of our method.

and it trains them so that the predicted latent $\hat{z}_{t+1}$ matches the encoded next frame $\text{enc}_\theta(x_{t+1}) = z_{t+1}$. This objective $\mathcal{L} = (z_{t+1} - \hat{z}_{t+1})^2$ alone has a trivial solution. The encoder can simply map every image input to a constant vector output, making the loss zero but removing any information about the environment and providing no benefit for dynamics. This failure is an example of the representation collapse that comes from the JEPA objective without the full pixel reconstruction loss. LeWM rules it out using SIGReg [20], which pushes the distribution of latents toward an isotropic Gaussian along random one-dimensional projections.

SMWM [7] removes SIGReg and regularizes with inverse dynamics to prevent collapse instead. A small MLP is given a pair of consecutive latents $[z_t, z_{t+1}]$ and predicts the action a_t that moved from z_t to z_{t+1} . Reducing this loss $\mathcal{L} = (\hat{a}_t - a_t)^2$ requires the latent space to not collapse, and it makes the latent space have action-relevant information while prescribing no distribution shape like SIGReg would.

Neither model structures the latent along the factorization of Sec. III-A. Both treat controllable and uncontrollable information equally, allowing for distractor signals to impact the representation and collapse the latent space.

C. CF-JEPA: Learning Controllability Factorized World Models

Where LeWM works well under nominal conditions across the tested tasks, it fails in the presence of visual disturbances. To address this, we modify the latent space formulation for increased robustness. Our insight is that we only need a certain amount of the information in the image in order to solve the control task, so we should arrange the learning objective with this as a goal. Our CF-JEPA explicitly factorizes the latent space for that goal, allowing us to use the control relevant latent subspace and ignore the other unnecessary visual information in a control irrelevant latent subspace.

Learning our world model proceeds similarly as in LeWM [6]. We assume access to an offline dataset $\mathcal{D}$ of state-action transitions on a task-by-task basis. Given this reward free data, we then train an encoder f which maps a given observation x_t to a latent vector z_t . Once we have the latent

vector, we train a separate predictor network which predicts the next latent vector given the current one and current action.

We structure our latent space into two regions, $z_t = [z_t^c, z_t^u]$, where z_t^c is a controllable region which should affect only the controllable part of the world and z_t^u is an uncontrollable region which should handle everything else and explicitly not model the control.

The learning objective of our model is as follows.

Forward Dynamics. First, we have the forward prediction loss mentioned above. We compute this forward loss on a per-subspace basis, where the controllable region only influences the controllable region, and likewise for the uncontrollable region. Also, only the controllable region uses the action, as the uncontrollable region by construction is not influenced by the action. Specifically, we have one predictor g that is action conditioned for z_t^c and another predictor h with no action conditioning for z_t^u . $g(a_t, z_t^c) = \hat{z}_{t+1}^c$, $h(z_t^u) = \hat{z}_{t+1}^u$. The two predictions are concatenated back into $\hat{z}_{t+1} = [\hat{z}_{t+1}^c, \hat{z}_{t+1}^u]$, and the loss is:

$$\mathcal{L}_{\text{fwd}} = (z_{t+1} - \hat{z}_{t+1})^2 \quad (3)$$

Inverse Dynamics. Next, we have an inverse dynamics loss. This is to set up the latent space controllability structure above and also to focus on the important aspect of the model, which is the ability to control downstream tasks. Specifically, we use another MLP predictor ψ that predicts the action given the current and next controllable latent, $\psi(z_t^c, z_{t+1}^c) = \hat{a}_t^c$.

$$\mathcal{L}_{\text{inv}} = (a_t - \hat{a}_t^c)^2 \quad (4)$$

Adversarial Dynamics. Then, we have an adversarial dynamics loss. This is to bolster the latent space structure by penalizing action predictability of the uncontrollable space. Here, we proceed similarly as with the inverse dynamics loss, using a MLP predictor ϕ to predict actions between the uncontrollable latents, but prepend a gradient reversal layer [39] to ϕ to actively penalize effective prediction, $\phi(z_t^u, z_{t+1}^u) = \hat{a}_t^u$.

$$\mathcal{L}_{\text{adv}} = (a_t - \hat{a}_t^u)^2 \quad (5)$$

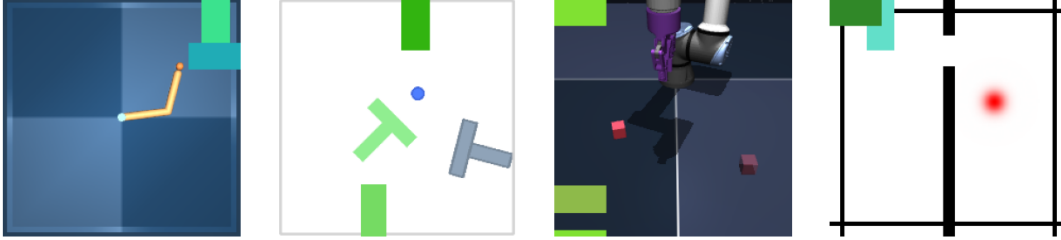


Fig. 3: Tasks with example distractors (N=2).

TABLE I: Model components and the loss weight that is tuned per task, λ for the inverse dynamics loss of SMWM and β for our adversarial loss. LeWM has neither, so no weight applies.

	LeWM	SMWM	CF-JEPA
<i>Model components</i>			
Forward loss	O	O	O
SIGReg	O	X	O
Inverse dynamics	X	O(full z)	O(z^c)
Adversarial dynamics	X	X	O(z^u)
Latent factorization	X	X	O
Control	z	z	z^c
<i>Per-task loss weight</i>			
Reacher	–	5	10
PushT	–	30	3
OGBench Cube	–	1	10
TwoRoom	–	0.1	3

SIGReg. Finally, we utilize the same SIGReg loss as in LeWM for preventing collapse of the latent space. A schematic can be seen in Fig. 2. The full objective is the following:

$$\mathcal{L} = \mathcal{L}_{\text{fwd}} + \alpha \mathcal{L}_{\text{inv}} + \beta \mathcal{L}_{\text{adv}} + \gamma \mathcal{L}_{\text{SIGReg}} \quad (6)$$

where for all tasks $\alpha = 1$ and $\gamma = 0.09$ while β was specified on a task by task basis. The lower block of Table I lists these per-task weights, alongside the weight λ that SMWM tunes for its own inverse dynamics loss.

Table I also summarizes how the three models differ. LeWM relies on the forward loss and SIGReg alone, SMWM replaces SIGReg with an inverse dynamics loss over the full latent, and we keep SIGReg while restricting the inverse dynamics loss to z^c and adding the adversarial loss on z^u . Also, since we factorize the control-relevant components explicitly, we use only z^c for control, while both others use the full latent z .

Control. For control, we plan using the world model in latent space. We optimize via model-predictive control using the cross entropy method [40]. Given a goal image, we compute the goal latent $z_{\text{goal}} = f(x_{\text{goal}})$ and compute the cost based on our distance in latent space from that goal, accumulating the cost over the predicted latent z_k^c at each step k in the receding horizon control.

TABLE II: Nominal task performance. We report the values for LeWM and SMWM directly from the source papers and list our values as mean $\pm$ standard deviation across three training seeds.

Task	LeWM	SMWM	CF-JEPA
Reacher	0.86	0.66	0.61 (± 0.03)
PushT	0.96	0.83	0.78 (± 0.04)
OGBench Cube	0.74	0.84	0.81 (± 0.03)
TwoRoom	0.87	0.99	0.95 (± 0.04)

$$C_{\text{CEM}} = \sum_{k=0}^K \|z_k^c - z_{\text{goal}}^c\| \quad (7)$$

D. Implementation Details

Our model architecture follows heavily from LeWM [6].

Encoder. The encoder f is a vision transformer, using the ViT-Tiny model from HuggingFace, with a patch size of 14. The output latent vector z is 192 dimensional, split so $z^c \in \mathbb{R}^{128}$ and $z^u \in \mathbb{R}^{64}$.

Forward model. The forward model consists of two transformer predictor networks, one action-conditioned for the controllable latent (g) and one action-free for the uncontrollable latent (h). The controllable network uses learned position embeddings and causal attention along with AdaLN-zero conditioning on the action to predict z_{t+1}^c given the input $[z_t^c, a_t]$. The action-free network uses a smaller transformer with causal attention to predict z_{t+1}^u given the input z_t^u .

Inverse model. The inverse model ψ consists of a two layer MLP network with GELU activations that maps $[z_t^c, z_{t+1}^c]$ to a_t .

Adversarial model. The adversarial model ϕ consists of a two layer MLP network with GELU activations augmented with a gradient reversal layer that maps $[z_t^u, z_{t+1}^u]$ to a_t .

Training. We train all models using AdamW with warmup using a learning rate of $5e-5$ and weight decay of $1e-3$. We train for 10 epochs on each task using a batch size of 128. All training was done using a machine with a single NVIDIA 5090 GPU and each world model took less than 8 hours to train.

Planning. For CEM planning, we consider 300 candidate actions per iteration and 30 iterations per planning session.

We plan out five actions and replan over a receding horizon of five.

Decoder. (Diagnostic Only) For latent space analysis after training, we also trained a decoder to map the latent vector z from the trained frozen encoder f to the pixel inputs x . The decoder is a stack of transposed convolutions that maps to the input image, in a manner similar to DCGAN [41].

IV. EXPERIMENTAL RESULTS

We evaluate our method to answer the following research questions:

- 1) How does CF-JEPA perform against the baseline methods under nominal conditions?
- 2) How does controllability factorization improve control performance under visual distractors?
- 3) Does the robustness of CF-JEPA and SMWM hold when the action space requires perceiving the state of the agent?
- 4) How does controllability factorization separate control-relevant information from distractors in the latent space?

A. Experimental Setup

We compare CF-JEPA against two baselines. LeWM [6] is the base model that our method builds on, and it shows strong planning performance across our tasks. SMWM [7], in contrast, makes its latent action-informative through an inverse dynamics loss, so we expect it to be more robust to visual disturbances which do not impact the action. Comparing against both lets us test whether controllability factorization adds robustness beyond these state-of-the-art models.

For the four tasks in Fig. 1, we use the datasets released with LeWM [6]. Each dataset was collected with an established policy for its task, ranging from expert demonstrations to scripted and learned controllers. Our method is agnostic to how the data is collected, and it only requires that the trajectories cover the dynamics of the environment (Sec. III-A). We train and evaluate all models with code adapted from [42], [43].

The datasets are as follows:

- 1) PushT, from [17]. We use 2,300,000 frames across 18,000 episodes collected from expert demonstrations.
- 2) OGBench Cube, from [44]. We use 2,000,000 frames across 10,000 episodes collected with a scripted policy.
- 3) TwoRoom, from [45]. We use 920,000 frames across 10,000 episodes of noisy scripted data.
- 4) Reacher, from [46]. We use 2,000,000 frames across 10,000 episodes collected using SAC [47].

B. Performance under Nominal Conditions

Under nominal conditions (Table II), CF-JEPA ranks second on OGBench Cube and TwoRoom, within 0.04 of the best model, and third on Reacher and PushT. On the latter two tasks, it trails LeWM by 0.25 and 0.18, respectively. Since we track very closely to SMWM, we hypothesize that this gap arises from the inverse dynamics objective shared by SMWM and CF-JEPA, which requires the latent to encode the dynamics of the agent in addition to predicting future states. This performance degradation is therefore expected, and the ranking changes substantially once visual distractors are introduced, as shown in Sec. IV-C.

TABLE III: Task performance under standard distractor settings ($N=2$) For all tasks, we report mean $\pm$ one standard deviation.

Task	LeWM	SMWM	CF-JEPA
Reacher	0.07 (± 0.05)	0.1 (± 0.03)	0.48 (± 0.12)
PushT	0.01 (± 0.01)	0.86 (± 0.02)	0.87 (± 0.03)
OGBench Cube	0.42 (± 0.02)	0.84 (± 0.01)	0.85 (± 0.02)
TwoRoom	0.33 (± 0.03)	0.98 (± 0.05)	0.95 (± 0.03)

C. Performance under Visual Perturbation

1) *Distractor Configuration:* To evaluate how our method captures the underlying controllable dynamics and ignores irrelevant features, we consider the case of perturbing the scene with additional visual distractors. Specifically, we add rectangles with a single randomly chosen color which are placed outward from a randomly chosen edge of the image frame. Each distractor is 49 pixels long perpendicular to the edge of the 224×224 image and the width is a random value chosen in the range [18, 36] pixels. By default, two rectangles are used, although that number N is a hyperparameter that we adjust later. Each rectangle moves along the image edge with a constant random velocity in the range [3, 18] pixels/step. The affected area of the frame is chosen such that the task is not occluded, so under ideal conditions, control performance should be unaffected. An example of each task with the distractor is shown in Fig. 3.

2) *Standard Distractor Performance:* To demonstrate the robustness of CF-JEPA to visual distractors, we evaluate all models on the four tasks under the standard distractor setting with $N = 2$. Table III shows the results across the different tasks for our method and the baselines. All models were trained with images augmented with $N=2$ distractors. The data is collected over three seeds. The baseline LeWM degrades on every task, and it collapses on Reacher and PushT, where its success rate falls below 0.1. SMWM performs much stronger, as the inverse dynamics loss also provides an amount of robustness, but also collapses under distractors in the Reacher task. Only our model did not collapse for any task, showing the benefit of the factorization for improving robustness against task-irrelevant distractors. CF-JEPA even performs better with distractors than under nominal conditions on PushT (0.87 vs. 0.78) and OGBench Cube (0.85 vs. 0.81). The OGBench results can likely be ascribed to the variance from the training seeds, but the PushT results are larger than that. Our hypothesis for this gap is that SIGReg forces both z^c and z^u to be populated and PushT under nominal conditions has no uncontrollable information, so some important information gets put into z^u , negatively impacting control. Under distractor conditions, this is not a problem as the distractor information can populate z_u . This is also not a problem for the other tasks as they include additional, uncontrollable background information that can get put into z^u .

For some quantitative support to this theory, we probed z^c for both the nominal and distractor trained models to see how well it encoded the proprioceptive state. We see a 0.07 ± 0.03 increase in R^2 for decoding the block angle and a 0.1 ± 0.02 increase for decoding the block position when going from the nominal to distractor case. This difference plausibly accounts for the control improvement under distractor conditions. An-

TABLE IV: Task performance as the number of distractors N increases. For each task, we report values up to the collapse of SMWM, except for OGBench Cube, where SMWM does not collapse.

Task	LeWM	SMWM	CF-JEPA
Reacher ($N = 2$)	0.08	0.1 (± 0.03)	0.48 (± 0.12)
PushT ($N = 2$)	0.01	0.86 (± 0.02)	0.87 (± 0.03)
PushT ($N = 16$)	–	0.36 (± 0.36)	0.76 ($\pm 0.$)
PushT ($N = 32$)	–	0.57 (± 0.01)	0.68 (± 0.06)
PushT ($N = 64$)	–	0.27 (± 0.25)	0.57 (± 0.01)
OGBench Cube ($N = 2$)	0.41	0.84 (± 0.01)	0.85 (± 0.02)
OGBench Cube ($N = 32$)	–	0.80 (± 0.02)	0.81 (± 0.01)
OGBench Cube ($N = 64$)	–	0.70 (± 0.05)	0.73 (± 0.01)
OGBench Cube ($N = 128$)	–	0.73 (± 0.05)	0.70 (± 0.02)
TwoRoom ($N = 2$)	0.34 (± 0.05)	0.98 (± 0.05)	0.95 (± 0.03)
TwoRoom ($N = 8$)	–	1 (± 0)	0.97 (± 0.01)
TwoRoom ($N = 16$)	–	0.6 (± 0.28)	0.94 (± 0)
TwoRoom ($N = 32$)	–	0.46 (± 0.12)	0.82 (± 0)

other nice benefit is that this distractor trained model can be deployed in the nominal environment and we still see better results, with a success rate of 0.87 ± 0.02 .

3) *Extreme Distractor Performance*: Since Reacher was the only task where SMWM failed, we did further examination to determine the cause. Our hypothesis is that Reacher is a more difficult task for a vision-based agent to complete as the arm is thin and takes up little of the screen, so there is little for the model to learn about for dynamics and little to help prevent collapse under the addition of visual perturbations. To further analyze this, we trained additional models with increased distractor number N in order to determine if the distractor tolerance was task-specific and Reacher was a particularly low robustness task. Table IV shows the performance per model across increased distractor values on a per task basis.

LeWM already fails at the smallest number of distractors, since nothing in its objective prevents the latent from collapsing under task-irrelevant visual signals. SMWM is more resilient, but its performance drops sharply as the distractors become dense. On PushT, its success rate falls from 0.86 at $N = 2$ to 0.27 at $N = 64$, while CF-JEPA more gradually degrades from 0.87 to 0.57 in the same range. On TwoRoom, SMWM falls from 0.98 at $N = 2$ to 0.46 at $N = 32$, while CF-JEPA only falls from 0.95 to 0.82. The high variances of SMWM shown here indicate a mixture of degrading performance akin to the CF-JEPA behavior on some seeds and complete latent collapse on others. CF-JEPA did not collapse in any of the configurations we tested.

Interestingly, OGBench Cube is the only task that shows a different trend, where SMWM does not collapse even under heavy visual disturbance and retains 0.73 at $N = 128$. For OGBench Cube, we tested up to distractor amounts of $N = 128$ before the exterior portion of the screen was entirely saturated with distractions and it still performed well. Even increasing the rectangle length, somewhat occluding the main robot and cube, still did not lead to SMWM collapse. We hypothesize that the specific task is robust due to its action space. Since it uses a differential IK solver for direct end effector control rather than something more low level like position control, all that is necessary for completing the control task is the goal latent.

In the extreme case, the control success remains unchanged

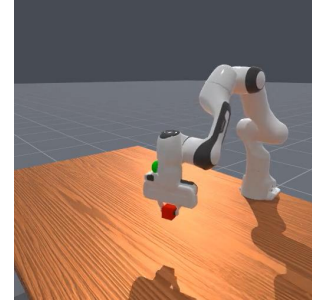


Fig. 4: Simplified ManiSkill Reach task used for further distractor evaluation. TABLE V: Task performance under $N=16$ distractor settings for ManiSkill Reach task. We report mean $\pm$ standard deviation across three seeds for each model.

Success Threshold (m)	LeWM	SMWM	CF-JEPA
0.06 (easiest)	0.55 (± 0.04)	0.79 (± 0.05)	0.83 (± 0.06)
0.05	0.51 (± 0.05)	0.62 (± 0.04)	0.72 (± 0.08)
0.04	0.44 (± 0.03)	0.58 (± 0.02)	0.64 (± 0.1)
0.03	0.36 (± 0.03)	0.38 (± 0.02)	0.53 (± 0.14)
0.025	0.32 (± 0.06)	0.30 (± 0.04)	0.48 (± 0.17)
0.02 (hardest)	0.21 (± 0.05)	0.22 (± 0.04)	0.35 (± 0.12)

even when the scene is fully occluded or the camera is moved so that the robot is out of frame. This is because the pixel observations are not used for control. Thus, the distractors have no effect on control performance unless there is encoder collapse to the point that the goal latent is meaningless, as in LeWM. Since this does not happen in SMWM or CF-JEPA, we see no failure.

D. Performance on Perception-Dependent Tasks

Because the action space of OGBench Cube bypasses the current observation, it cannot reveal whether a model perceives the agent robustly. We therefore evaluate on a task whose action space requires perceiving the state of the agent. We use a simplified variant of the ManiSkill [48] PickCube task (Fig. 4), in which the robot moves an already grasped cube to a goal position. Unlike OGBench Cube, the task uses joint position control, so the agent must infer the arm configuration from the image. We train all models on 250,000 frames and evaluate them with $N = 16$ distractors. Table V reports the success rate for different thresholds on the distance to the goal. At a loose threshold of 0.06 m, CF-JEPA and SMWM perform similarly, with a gap of only 0.04. As the threshold tightens, the gap widens to 0.15 at 0.03 m and 0.13 at 0.02 m. SMWM, which was as robust as CF-JEPA on OGBench Cube, falls behind once the task demands more precise perception.

E. Latent Space Analysis

We also performed analysis of the latent space generated by our model in comparison to the baselines. We trained a separate decoder network that takes as input the latent vector output from the frozen encoder network from each model with the goal of reconstructing the pixels from the original image. For a qualitative analysis, we look at the Reacher task. All models are trained with distractors, but we visualize the reconstructions on test frames without distractors. CF-JEPA learns to distinguish controllable information from uncontrollable information, as the 2nd and 3rd rows of Fig. 5

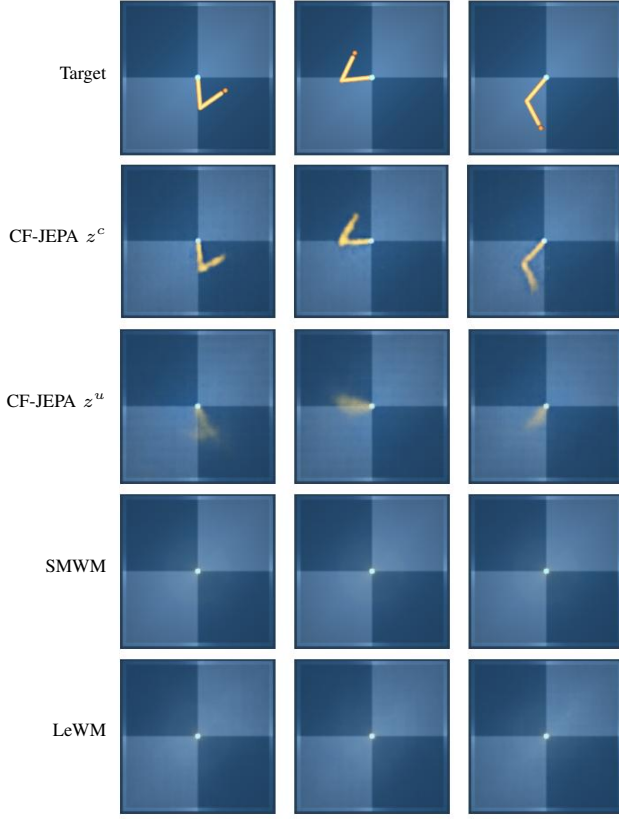


Fig. 5: Only the controllable latent of CF-JEPA retains the agent. Reconstructions are decoded from frozen encoders trained with distractors and shown on Reacher test frames without distractors. Compared with the target (1st row), the controllable latent z^c of CF-JEPA reconstructs the arm (2nd row), whereas the uncontrollable latent z^u only blurs it (3rd row). SMWM and LeWM lose the arm entirely (4th and 5th rows).

show qualitatively. It is clear from the 2nd row that the arm is well reconstructed, ignoring the extraneous distractor information to focus on the important information for control. On the other hand, CF-JEPA with z^u (the 3rd row) is incapable of reconstructing the Reacher arm well, as this latent subspace cannot represent the controllable information. Looking at the baselines, the 4th and 5th rows show that under distractor conditions, LeWM and SMWM are collapsed and entirely incapable of representing the Reacher agent.

As a quantitative measure consistent with the qualitative images in Figure 5, we use the participation ratio [49], defined as:

$$\text{PR} = \frac{\left(\sum_{i=1}^d \lambda_i\right)^2}{\sum_{i=1}^d \lambda_i^2}, \quad (8)$$

where the eigenvalues λ are computed from a random 256 frames collected from the entire dataset, to demonstrate the spread of variance in the latent space for the clean and distracted settings. Table VI shows this metric for each model on the Reacher task. From this, we can see that the distractors cause a large decrease for all models, with total collapse for SMWM and large decreases for LeWM and CF-JEPA. SMWM starts at a lower number as it has no SIGReg to spread out the latent space and falls to the minimum value of 1 due to collapse. Neither LeWM or CF-JEPA fall to zero due to SIGReg, but since LeWM falls 21.5 times compared to the

TABLE VI: Latent vector participation ratio under nominal and distracted conditions for the Reacher task. LeWM and SMWM both collapse much further than CF-JEPA, correlating with the worse decoder and control performance.

	LeWM	SMWM	CF-JEPA
Nominal	48.2 (± 0.8)	3.71 (± 0.07)	42.3 (± 0.1)
Distractor (N=2)	2.24 (± 0.23)	1. (± 0.01)	12.9 (± 0.35)

3.3 times for CF-JEPA, it is not performant for control or decoding, leaving CF-JEPA the only functional model tested.

V. CONCLUSION

This paper presents a method for training a JEPA world model with improved robustness to visual disturbances. We build our CF-JEPA off of LeWM, incorporating additional structure in the latent space via adding inverse dynamics losses so that we can factorize the space into controllable, action-conditioned, and uncontrollable, action-independent, spaces. We train our model for a set of four tasks under nominal conditions and under conditions with task-irrelevant distractor signals. Our model shows comparable performance against the baselines under nominal conditions, and improved performance under distracted conditions, being the only model tested that did not experience latent collapse.

Our experiments suggest two directions for future work. First, we validate CF-JEPA in simulation with a single family of distractors, and applying it to a real robot would test the factorization under richer visual conditions. Second, we assume that the uncontrollable part of the scene never affects the controllable part, so settings where the two interact, such as a person working next to the robot, would require extending the model.

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